

Build Scalable Backend with Kubernetes

Build and deploy a scalable backend service using Kubernetes with high availability and auto-scaling on a cloud provider.

Requirements

1. **Backend API** : Create a simple CRUD API (e.g., Posts)
2. **Docker** : Dockerize the application and Push to Docker Hub
3. **Cloud Deployment** : Deploy on either:GCP or AWS
4. **Kubernetes Setup** using Kubernetes Deploy backend using Deployment :
`minReplicas: 1 and maxReplicas: 5`
5. **Auto Scaling** : Implement **HPA and** Scale based on CPU usage 70%
6. **Database** using MongoDB : Deploy as **Replica Set (3 nodes)** , Use StatefulSet
7. **Service & Ingress** : Expose backend using Service (ClusterIP) , Expose API externally using Ingress

Validation (Must Demonstrate)

- API is accessible via Ingress
- Auto-scaling works (pods increase under load)
- High availability: Delete a backend pod → system still works , Delete a Mongo pod → data is not lost

Deliverables

1. **GitHub Repository contains** Source code , Dockerfile , Kubernetes manifests
2. **README** Setup steps , API endpoints
3. **Recorded Video (clear voice Arabic / English)**
Must include:
 - Architecture explanation
 - Running system demo
 - Auto-scaling in action
 - Pod deletion test (failover)

Acceptance Criteria

- Fully working deployment on cloud
- Proper Kubernetes usage (Deployment, HPA, Ingress)
- MongoDB Replica Set configured correctly
- No data loss after pod restart
- Clear explanation in the video